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Experience

Ginkgo Bioworks: Biosecurity
Boston, Massachusetts

Staff Software Engineer
March 2021 - November 2025

Tech lead for a HIPAA-compliant pathogen surveillance platform - COVID-19 in thousands of K-12 schools, with lab partners in 18 states, then multipathogen surveillance at 11 international airports (126 origin countries) - processing 15M+ samples and underpinning $733M in revenue.

- Led a multi-year platform evolution from single-pathogen COVID testing to multipathogen biosurveillance - refactoring live production systems while preserving historical data, existing visualizations, and CDC reporting pipelines.
- Decomposed the core monolithic data model into three normalized tiers - enabling complex multi-step, multi-result testing workflows.
- Led a team of ~11 as interim team lead through a period of rapid organizational change - running ceremonies, managing stakeholders, and keeping engineers focused on delivery; transitioned to a dedicated tech lead role once a manager was hired.
- Built the team's release process (code-freeze automation, release-engineer rotation), on-call, and a DR exercise; mentored junior engineers.
- DevOps: rebuilt CI/CD pipelines around Docker with GitHub Actions, codified infrastructure in Terraform (Auth0, AWS, Aptible, supporting services), migrated off legacy vendor systems, and achieved 8x build performance on self-hosted runners.
- Designed PHI removal from the platform as the business shifted from individual diagnostics to aggregate biosurveillance - balancing regulatory compliance, data retention, and system simplification.

Flask · SQLAlchemy · PostgreSQL · Terraform · GitHub Actions · Docker · Nix · AWS EC2/S3/CloudFront · Auth0 · Aptible · Grafana · New Relic · Cursor

Oliver Wyman: Digital
Boston, Massachusetts

Senior Software Engineer
October 2017 - February 2021

- DevOps: wrote idiomatic containers, stood up CI/CD into Kubernetes across Digital's project portfolio.
- Webservices: REST and GraphQL backends for analytics; built a system cataloguing all of Digital's projects and assets for InfoSec auditability.
- Data infrastructure: ETL and workflow tooling. Built out a schema migration framework for a major grocery retailer, eliminating downtime.
- Sustaining: led long-term operations for five projects in the mature phase of their lifecycle.

Python · Kubernetes & Kustomize · Jenkins Pipelines · Prefect · Node.js (Express & Vue) · Alembic · MS SQL Server · PostgreSQL · GraphQL

Lockheed Martin: Space Systems
Littleton, Colorado

Software Engineer - Orion: NASA's next crewed spacecraft
September 2014 - September 2017

- Principal architect in moving Orion ground software from pen and paper to full test automation, empowering NASA requirement verification.
- Built a Python/LaTeX-based engine for creating NASA deliverables, cutting months of paperwork for engineers (per deliverable) to a day.
- Subject-matter expert on the flight software verification automation tiger team, sharing processes proven on ground software.

Education

Colorado School of Mines B.S. Engineering, Mechanical Emphasis – CS & Economics minors
December 2013

Capstone: NASA Lunabotics Competition: built a regolith mining robot and competed with international teams at the Kennedy Space Center.

Projects

reelbarrow & AI-Assisted Engineering
2025 - Present

- Designed a review workflow for agentic development: a spec per project, a pull request reviewed thread by thread, and offline checks every change must pass. 4 pull requests merged upstream (3 to nixpkgs).
- Diagnosed passthrough of a Blu-ray drive across three virtualization layers (Proxmox hypervisor, Talos VM, Kubernetes device plugin & pod), then specified a phased ripping pipeline and verified its CD, DVD and Blu-ray edge cases, with an agent implementing each phase.
- Building a service that files ripped discs into a video library, using an LLM (Claude) to match Whisper transcripts to episodes. Each automatic filing needs a second, independent signal, because the model can be consistently and confidently wrong.

Pi agent harness · Claude Code · Claude API · Whisper · Local LLMs (Ollama) · FastAPI

Private Infrastructure Cloud
2015 - Present

- Architected a GitOps-driven platform from bare metal through immutable Kubernetes, hosting 24+ services, with archive-class file integrity.
- Redesigned backups after an audit found four silent failures (offsite replication had failed for seven months): a phased 3-2-1-0 spec, append-only restic, pull-only replication, per-dataset freshness checks. Pruned 30,404 snapshots to 1,083, reclaiming 2.32 TiB.

OpenTofu · Prometheus/Loki · Flux CD · Nix · SOPS secrets · ZFS · TLS · Talos Linux · Proxmox · CoreDNS · Tailscale · OpenEBS